Jaeyoung Yoon, Ph.D.

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EMPLOYMENT

Harvard Medical School Instructor, Department of Neurology	2026 – present
Boston Children's Hospital Scientist, F.M. Kirby Neurobiology Center	2023 – present
Massachusetts Institute of Technology Postdoctoral Fellow, McGovern Institute for Brain Research	2019 – 2023
Seoul National University Postdoctoral Associate, Medical Research Center	2019 – 2019

EDUCATION

Columbia University J.D. Candidate, Columbia Law School	2027 – 2030 (expected)
Ph.D., Seoul National University School of Biological Sciences, College of Natural Sciences / Department of Physiology, College of Medicine (*joint affiliation*) Thesis: *"Short-term synaptic plasticity and persistent activity in the prefrontal cortex"* (2016 – 2019: Military service, Republic of Korea Army)	2013 – 2019
B.S., Seoul National University Biological Sciences, College of Natural Sciences	2009 – 2013

AWARDS AND HONORS

Travel/Research Award, Mind-Brain-Behavior Initiative, Harvard University (2.0 k USD)	2024
Best Presenter Award, F.M. Kirby Neurobiology Center, Boston Children's Hospital (BCH)	2024
Y. Eva Tan Fellowship, Yang-Tan Center for Molecular Therapeutics in Neuroscience, Massachusetts Institute of Technology (MIT) (130.0 k USD)	2021 – 2023
Molecular Therapeutics Impact Report 2020 – 2022, MIT (featured)	2022
BK21 / BK21+ Fellow, National Research Foundation of Korea (NRF) (21.3 k USD)	2013 – 2017
Lecture and Research Scholarship & Merit-based Scholarships, Seoul National University (SNU)	2013 – 2014
Superior Academic Performance Scholarships, SNU	2009 – 2011

SELECTED PUBLICATIONS

Yoon J*, Terauchi A, Castro A, Stone S, Lidov H, Cohen A, Umemori H, Ferguson B*. (2026).
Human cortical and thalamocortical synaptic dysconnectivity in ASD. (*in preparation*) **(* co-corresponding author)**

Yoon J*, Silva R, Stone S, Lidov H, Osterweil E, Ferguson BR*. (2026).
High-frequency, high-intensity electrical stimulation selectively activates human fast-spiking interneurons.
Frontiers in Synaptic Neuroscience. DOI: 10.3389/fnsyn.2026.1843277 **(* co-corresponding author)**

Cho E, Kwon J, Lee G, Shin J, Lee H, Lee SH, Chung CK*, **Yoon J***, Ho WK*. (2024).
Net synaptic drive of fast-spiking interneurons is inverted towards inhibition in human FCD I epilepsy.
Nature Communications. DOI: 10.1038/s41467-024-51065-7 **(* co-corresponding author)**

Yoon J. (2024). Geometrical determinant of nonlinear synaptic integration in human cortical neurons.
arXiv. DOI: 10.48550/arXiv.2408.05633

Yoon JY, Lee HR, Ho WK, Lee SH. (2020). Disparities in short-term depression among prefrontal cortex synapses sustain persistent activity in a balanced network. *Cerebral Cortex*. DOI: 10.1093/cercor/bhz076

OTHER PUBLICATIONS

Singh A, et al. (2026). FMR1 reduction alters cellular and circuit properties in human cortex. (*under review at Science*)

He M, et al. (2026). Human cortical neuron-based assays to study axon regeneration. (*in preparation*)

Yoon J. (2025a). Acute human brain slice preparation for ex vivo electrophysiology and imaging. *protocols.io.* DOI: 10.17504/protocols.io.ewov124qogr2/v1

Yoon J. (2025b). Pulse splitter for 2-photon glutamate uncaging and ex vivo electrophysiology. *protocols.io.* DOI: 10.17504/protocols.io.eq2lyxpjwgx9/v1

Lee BJ, et al. (2025). A specific association of presynaptic K^+ channels with Ca^{2+} channels underlies K^+ channel-mediated regulation of glutamate release. (*under review at Proceedings of the National Academy of Sciences of the USA*)

Yoon JY, Choi S. (2017). Evidence for presynaptically silent synapses in the immature hippocampus. *Biochemical and Biophysical Research Communications*. DOI: 10.1016/j.bbrc.2016.12.044

INVITED TALKS

"Human cortical and thalamocortical synaptic dysconnectivity in ASD". *Gordon Research Conference.* Barga (LU), Italy. (Feb 2026)

"Human cortical and thalamocortical synaptic dysconnectivity in ASD". *Department of Neurosurgery, Yonsei University College of Medicine.* Seoul, Korea. (Dec 2025)

"Electrophysiological hallmarks of epilepsy and autism in the human neocortex". *China-Japan-Korea Neuroscience Meeting.* Incheon, Korea. (Aug 2025)

"Thalamocortical synaptic hyperconnectivity in the human neocortex with ASD". *F.M. Kirby Neurobiology Center, BCH & Harvard Medical School (HMS).* Boston, MA, USA. (May 2025)

"Net synaptic drive of fast-spiking interneurons is inverted towards inhibition in human FCD I epilepsy". *New England Bioscience Society*. Boston, MA, USA. (Sep 2024)

"Synaptic integration in human neuronal dendrites". *Department of Physiology, SNU College of Medicine*. Seoul, Korea. (Dec 2023)

"Subcellular connectivity and synaptic integration in cortical pyramidal neurons". *Yang-Tan Center for Molecular Therapeutics in Neuroscience & Tan-Yang Center for Autism Research Joint Symposium, McGovern Institute for Brain Research (MIBR), MIT.* Cambridge, MA, USA. (Jul 2021)

MEETING ABSTRACTS

Yoon J*, Terauchi A, Castro A, Stone S, Lidov H, Cohen A, Umemori H, Ferguson B* **(* co-corresponding author).** "Human cortical and thalamocortical synaptic dysconnectivity in ASD". *Federation of European Neuroscience Societies (FENS) Forum.* Barcelona, Spain. (Jul 2026)

Yoon J*, Terauchi A, Castro A, Cohen A, Umemori H, Ferguson B* **(* co-corresponding author).** "Human cortical and thalamocortical synaptic dysconnectivity in ASD". *Gordon Research Conference.* Barga (LU), Italy. (Feb 2026)

Yoon J*, Ferguson B* **(* co-corresponding author).** "Functional hyperconnectivity of thalamocortical synapses in human ASD". *Harvard Mind-Brain-Behavior Research Showcase.* Cambridge, MA, USA. (Apr 2025)

Yoon J*, Ferguson B* **(* co-corresponding author).** "Functional hyperconnectivity of thalamocortical synapses in human ASD". *F.M. Kirby Neurobiology Center Retreat.* Boston, MA, USA. (Mar 2025)

He M, Chung L, Hathaway D, **Yoon J**, Yan T, Yu S, Chalif J, Ferguson B, Osterweil E, He Z. "Human cortical neuron-based assays to study axon regeneration". *F.M. Kirby Neurobiology Center Retreat.* Boston, MA, USA. (Mar 2025)

Cho E, Kwon J, Lee G, Shin J, Lee H, Lee SH, Chung CK*, **Yoon J***, Ho WK* **(* co-corresponding author).** "Net synaptic drive of fast-spiking interneurons is inverted towards inhibition in human FCD I epilepsy". *Gordon Research Conference.* Waterville Valley, NH, USA. (Aug 2024)

Yoon JY, Lee HR, Ho WK, Lee SH. "Disparities in short-term depression among prefrontal cortex synapses sustain persistent activity in a balanced network". *Neuro2019.* Niigata, Japan. (Jul 2019)

Yang CH, **Yoon JY**, Ho WK, Lee SH. "Presynaptic mitochondrial calcium release during high-frequency train pulse enhances short-term facilitation". *Korean Society for Brain and Neural Science (KSBNS) Annual Meeting*. Goyang, Korea. (Oct 2016)

TECHNICAL EXPERIENCE

***ex vivo* electrophysiology (patch clamp) in acute human brain slices:** 2019 – present

- Whole-cell (somatic, dendritic, & paired) or excised (outside-out, nucleated, inside-out) patch clamp in human neocortex (PN, FSIN, nFSIN; in L2/3, L5, L6; from temporal, frontal, occipital, parietal, insular lobe)
- Human brain slice preparation, from > 100 adult and pediatric patients diagnosed with tumor or epilepsy; healthy and patched at soma and distal apical dendrite up to 132 h post-resection
- Research Non-Employee Collaborator, Massachusetts General Hospital (MGH) (2021 – 2023)

***ex vivo* electrophysiology (patch clamp) in other applications:** 2013 – present

- Slice electrophysiology setup at BCH/HMS (CLS 13052), MIT MIBR (46-6178), SNU College of Medicine (2-726, 2-724), and SNU College of Natural Sciences (504-201)
- Patch clamp in rodent brain slices (mouse, rat); in neocortex (L2/3, L5, L4, L6; TeA, PFC, ACC, V1, S1, RSC), hippocampus (CA1, CA3, DG), thalamus (MD), amygdala (BLA), and Calyx of Held
- Patch clamp in human brain slice culture & human cortical organoids
- Single-cell RNA (scRNA) sequencing with patch clamp (Patch-seq) from human neurons for transcriptomic analysis
- Computational modeling of vesicular dynamics and network biophysics, with optogenetic or electric stimulation under physiological or therapeutic scenarios
- Electrophysiological characterization of connexin mutants expressed in human embryonic kidney cells (HEK 293) for therapeutic applications (in collaboration with The Collective for Psychiatric Neuroengineering, Duke University & Howard Hughes Medical Institute (HHMI); 2025)
- Local field potential (LFP) recording in acute brain slices

2-photon excitation microscopy (2PEF) and fluorescence microscopy: 2013 – present

- MIT MIBR 2-photon core facility (46-6178) setup and management, including user training (6 postdocs from MIT & Broad Institute of MIT and Harvard trained during 2019 – 2023)
- 2-photon glutamate uncaging (2PGU), with 8x pulse splitter for enhanced 2PEF; including setup and application (provided to University of Ottawa in 2022)
- Intracellular calcium imaging & structural imaging for functional and morphological analysis
- Subcellular channelrhodopsin-assisted circuit mapping (sCRACM)

Data analysis and processing: 2013 – present

- Software development for electrophysiology and 2-photon imaging data analysis (https://github.com/flosfor/pvbs)
- Contribution to abfload: MATLAB function for processing .abf file format (https://github.com/fcollman/abfload)

Teaching experience: 2013 – present

- Instructor in Neurology, HMS (2026 – present)
- Mentor, Postdoctoral Fellow-Student Mentorship Program, BCH (for socially underprivileged students) (2025 – 2026)
- Teaching Assistant, Data Analysis in Neuroscience Workshop, Interdisciplinary Program in Neuroscience, SNU (2018)
- Teaching Assistant, Biology Lab 1 & 2 / Biological Sciences Research Lab 1 & 2, SNU (2013 – 2014)
- Laboratory personnel training (SNU, MIT, BCH, HMS) (2013 – present)

Other research and laboratory experience: 2013 – present

- *ad hoc* reviewer for *Nature Communications*, *Neuron*, *Cell Reports*, and *Frontiers in Synaptic Neuroscience*
- Technical assistance for *in vivo* patch clamp and Neuropixels setup at MIT MIBR (46-6171)
- Experience with plasmid DNA purification, viral vector packaging, immunohistochemistry (IHC), stereotaxic surgery, and expansion microscopy (ExM)

LANGUAGES

English (bilingual), Korean (bilingual), Italian (proficient), French (intermediate), German (beginner), MATLAB (proficient)

MEMBERSHIPS

Society for Neuroscience (SfN), Federation of European Neuroscience Societies (FENS), Italian Society for Neuroscience (SINS), Korean Physiological Society, Korean Society for Brain and Neural Sciences (KSBNS), Japan Neuroscience Society, US Chess Federation (USCF; chess.com rating ≤ 2131)